

SAPTARSHI MANNA

Full Stack Developer | MERN Stack | Data Structures & Algorithms

West Bengal, India

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Education

Jadavpur University

B.Tech in *Information Technology* | CGPA: 7.5+/10

Nov 2024 – 2028 (Expected)

West Bengal, India

Technical Skills

- **Languages:** C, C++, Java, JavaScript, Python
- **Core CS:** Data Structures & Algorithms, Dynamic Programming, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks
- **Frontend:** React.js, HTML, CSS, Tailwind CSS, Responsive Web Design, State Management
- **Backend:** Node.js, Express.js, REST API Development, Authentication, Authorization, JWT, MVC Architecture
- **Databases:** MongoDB, MySQL, Database Design, CRUD Operations, Indexing, Query Optimization
- **Tools:** Git, GitHub, Linux, Postman, API Integration

Projects

Real-Time Messaging Platform [🔗](#) | MERN Stack, Socket.IO, JWT

May 2026

- Engineered a **real-time messaging platform** using React, Node.js, Express.js, MongoDB, and Socket.IO, leveraging **WebSocket communication, event-driven architecture, and real-time messaging** for sub-second message delivery.
- Implemented **authentication and authorization** using JWT, along with chat history persistence and online/offline presence tracking to enhance security and user experience.
- Developed scalable **REST APIs**, WebSocket event handlers, and concurrent connection management to support low-latency message synchronization.
- Designed MongoDB schemas, CRUD operations, and optimized database queries for efficient storage and retrieval of user and chat data.
- Deployed the application on **Vercel** (Frontend) and **Render** (Backend), configuring production-ready environment variables and third-party integrations.

Real-Time Collaborative Coding Interview Platform [🔗](#) | MERN Stack, Clerk, Stream, Inngest

Jun 2026

- Engineered a **real-time coding interview platform** featuring **collaborative editing, video conferencing, live chat, and real-time communication** for remote technical assessments.
- Designed secure interview workflows using Clerk authentication, room locking, participant management, and **Role-Based Access Control (RBAC)**.
- Integrated a **code execution engine** supporting multiple programming languages with automated test-case validation and instant feedback.
- Developed interactive dashboards with **live interview statistics, notifications, and session tracking** to improve user engagement and monitoring.
- Optimized application performance through **TanStack Query caching, Inngest background job processing, API integration, and scalable REST API development**, reducing redundant network requests and improving responsiveness.

Achievements

- Secured **WBJEE Rank 505** among **1,40,000+** candidates in 2024.
- Solved **200+** problems on **Leetcode** and **Codforces** — [saptarshi.codes](#)
- Achieved **Codeforces Pupil** (1200+) — [Saptarshi.codes](#)